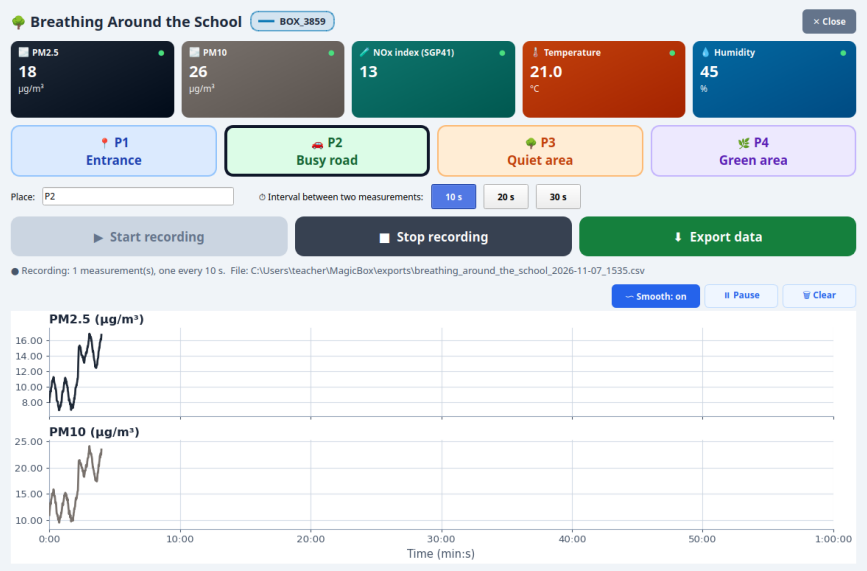
Breathing Around the School

One box carried to four places, and the question of where the school's air is most loaded

Level: Explorer and Scientist — ages 11–19 (Romanian school years V–XII) • **Time in class:** one 50-minute lesson outdoors; a two-lesson project with the analysis in Excel • **Sensors:** BMV080 (particles), SGP41 (NOx index), BME280 (temperature, humidity) — any one of them



The measuring page, with a box carrying all three sensors. The values in the picture are simulated; your own will differ.

Before you start

Who it is for

Explorer and Scientist — ages 11–19 (Romanian school years V–XII). In years V–VI pupils compare the places and say which was warmest and which was dustiest; from year VII they add where the particles and the nitrogen oxides come from (geography, chemistry), and at upper secondary level the mean, the spread within one place and the limits of a five-minute sample.

Assumed knowledge

Reading a value from a screen, finding a place on a map of the school, and the mean of a few numbers. Nothing about particles or nitrogen oxides: the activity introduces them.

Time in class

One 50-minute lesson: the prediction and the route in class (10), four places of about seven minutes each (30), the first comparison back in class (10). Project: a second lesson for the means in Excel and the discussion.

Teacher preparation

20 min — switch the box on at least ten minutes beforehand: the NOx index needs about a minute before its first value and the particles 15 seconds. **Check the sensors on the start page**: each one the box reports turns green and reads **The box sees it**. Charge the laptop. Walk the route once and fix the four places on a printed map of the school, **P2 behind the fence or on the pavement, several metres from the kerb**.

Sensor settling time Particles: one value a second, from 15 seconds after power-up. NOx index: about a minute after power-up. Temperature and humidity follow a move from indoors to outdoors, or from sun to shade, only after a few minutes, because the box itself has to warm or cool. **At each place, wait one to two minutes before recording.**

Sensors and boxes The **BMV080** for PM2.5 and PM10, the **SGP41** for the NOx index, the **BME280** for temperature and humidity, on a box with firmware 2.5.0 or newer. The lesson works with any one of them; the start page says **Missing sensors leave their columns empty in the file. You can start anyway. Classroom boxes on firmware 2.3.0 or 2.2.6, with only the usual sensors**, give temperature and humidity alone; for them *Urban Heat Islands* is the better lesson.

Three routes

Short One 50-minute lesson: the prediction and the route (10), the four places P1 to P4 with five minutes of recording at each and the walk between them (30), the first comparison back in class (10).

Full Short version, 30 min: two places only, the busy road (P2) and the green area (P4), five minutes each. The contrast is usually clearest between these two.

Project Two-lesson project: the walk as above, then a lesson in which pupils open the export in Excel, compute the mean at each place and set it beside the official station nearest the school. A second walk at another time of day (when parents drop children off, at midday) makes the best repeat.

The application starts in English. Buttons are named here exactly as they appear on the screen.

If the lesson is not on the Activities page

The lesson needs sensors that the usual classroom boxes do not have, so on some computers the app hides it. Open **Settings**, go to the **Visible activities** section and tick *Breathing Around the School*; the lesson appears at once on the **Activities** page. The tick stays on that computer.

1. Where this fits in the curriculum

- **Geography** — in year V, the composition of air and reading a map; in year IX, air pollution and its sources. Here pupils map a pollutant on the few hundred metres around their own school.
- **Biology** — in year VII, human respiration; at upper secondary level, the effect of fine particles and of nitrogen dioxide on the airways.
- **Physics** — in years VI–VII, heat and its transfer: asphalt in the sun warms the air above it, while grass and trees shade the ground and evaporate water. In year VIII, the scattering of light, which is how the particle sensor counts dust.
- **Mathematics** — in years VI–VII, the arithmetic mean; at upper secondary level, comparing two means against the spread of the readings behind them.

The core skill is **comparing places with a quantity that changes from one second to the next**. A single reading at the road says little; the mean of thirty readings, set against the mean at the green area, says more.

2. Time plan

Route	Time	What happens
One lesson	**50 min**	The prediction and the route on the map (10), the four places with five minutes of recording at each (30), the first comparison in class (10).
Short	30 min	Only P2 and P4, five minutes each, and a quick comparison of the cards.
Lesson 2	50 min	The export opened in Excel: the mean at each place, the largest and smallest value, and the official station's value for the same hour.
A second walk	30 min	The same route at another time of day or in other weather, to see which differences come back.

3. Materials

Item	Per group	Notes
MagicBox with at least one of the BMV080, SGP41, BME280	1	The box has three sockets; these three sensors fill them.
Laptop with the app, battery charged	1	Carried with the box on its USB cable. Over WiFi the box must stay within reach of the network.
A printed map of the school	1	With P1 to P4 marked and the route between them.
Stopwatch or phone	1	For the five minutes at each place and to note the time of events (a bus passing, a mower).
The pupil sheet on a clipboard	1 per pupil	Outdoors there is no desk.

Preparation the day before. Choose the four places so that each stands for something: **P1 Entrance**, where pupils and parents arrive; **P2 Busy road**, beside the busiest street, behind the fence; **P3 Quiet area**, a corner of the yard away from cars; **P4 Green area**, on grass or under trees. Time the walk between them. Indoors, hold your hand about ten centimetres above the particle sensor: the yellow warning should appear on the measuring page within a few seconds and go away when you lift your hand. That tells you pupils will see it too.

4. Scientific background for the teacher

Three sensors, three questions about the air outside

Particles (BMV080). PM2.5 and PM10 are the mass of particles smaller than 2.5 and 10 micrometres in a cubic metre of air, in $\mu g/m^3$. Around a school they come from exhaust, from brakes and tyres, from road dust lifted by traffic, from stoves and boilers in the heating season, from building sites and from the yard itself. PM2.5 is mostly smoke and fine combustion particles; they stay in the air for days and drift across a whole town, so much of the PM2.5 at any school is the town's background, and two places a hundred metres apart often differ little. PM10 also holds coarser dust, which settles within metres of where it was raised: near a dusty road or a yard full of running children it can differ much more. The sensor has no fan. It watches, with a laser, the air about 5 mm above its lens, counts the specks that cross the beam and turns the count into a mass. In fog or very humid air the

particles take up water and look larger, so readings rise.
Nitrogen oxides (SGP41). NO and NO_2 form when fuel burns hot, mostly in engines (diesel ones above all) and in boilers. Along a busy road the NO_2 is higher than a little further away, and it falls off over the first tens to hundreds of metres. The SGP41 does not measure a concentration. It gives Sensirion's **NOx index**: 1 means the air as clean of nitrogen oxides as the sensor has seen it lately, and a higher number means more than that. A bus pulling away beside the box can lift the index for a few seconds; in a quiet corner it usually stays at 1.
Temperature and humidity (BME280). Asphalt and concrete absorb most of the sunlight that falls on them and warm the air just above. Grass and trees shade the ground and evaporate water through their leaves, so the air there is often cooler and a little more humid. On a sunny day the difference at a metre above the ground is commonly one to a few degrees; under a grey sky it may vanish. A box held in direct sun heats up and reads too warm.

4.1 The thresholds and what they mean

Quantity	Reference	What it means
PM2.5	WHO: 15 $\mu g/m^3$ as a 24-hour mean	The World Health Organization's 2021 guideline. The recommended annual mean is 5 $\mu g/m^3$.
PM10	WHO: 45 $\mu g/m^3$ as a 24-hour mean	The same guideline; the recommended annual mean is 15 $\mu g/m^3$.
NOx index (SGP41)	1 = no more nitrogen oxides than usual	An index, not a concentration. It cannot be set beside the WHO guideline for NO_2.
Temperature, humidity	no threshold	Compare the places with one another, measured at the same height and in the shade.

Be careful with pupils on one point: **the WHO values are averages over a whole day**. Five minutes at 25 $\mu g/m^3$ beside the road do not "exceed the guideline"; they show that the air was more loaded there during those minutes. On a walk the fair comparison is between the places, measured the same way within the same half hour.

5. Lesson flow

The activity opens on a start page with the three sensors the lesson asks for, each with a picture: green with **The box sees it** when the box reports it, red with **Not seen by the box** when it does not. **Start** opens the measuring page, where the walk takes place.

1. **The prediction and the route (10 min).** Show the map with P1 to P4. Each pupil writes down which place will have the most particles, which the highest NOx index, and which will be warmest, with one reason. Most classes choose the road for everything; the data do not always agree.
2. **The start page (2 min).** Open the activity from the **Activities** page, check that the tiles are green and press **Start**. A sensor plugged in late appears after the box is unplugged and plugged in again; the app can stay open.

Technical note — what the screen shows and what is saved

The cards and the chart. Five cards: PM2.5 and PM10 in $\mu g/m^3$, **NOx index (SGP41)** without a unit, **Temperature** in °C and **Humidity** in %. The chart below them draws PM2.5 and PM10 over the last hour.

The place and the interval. Four large buttons, **P1 Entrance**, **P2 Busy road**, **P3 Quiet area** and **P4 Green area**, fill the **Place:** field with their code. For a fifth place, type its name in the field instead. The buttons **10 s**, **20 s** and **30 s** next to **Interval between two measurements:** set how often a row is recorded; 10 s gives thirty rows in five minutes.
Recording. Start recording works only once a place is chosen; otherwise the app asks **Choose the place first (P1 to P4) or type its name. Stop recording** ends it. The line under the buttons counts the rows and names the file.
The file. Every row is written at once to `MagicBox\exports\breathing_around_the_school_<YYYY-MM-DD_HHMM>.csv`, so nothing is lost if the laptop shuts down. **Export data** saves the session as an Excel workbook (`.xlsx`, with the sheets `Data` and `Legend`) or as a CSV. The column names are in English whatever the app's language:

`Place`	P1 to P4, or the name typed; the `Legend` sheet names the four codes
`Timestamp`, `Elapsed (s)`	date and time of the row; seconds since the recording started
`PM2.5 (µg/m³)`, `PM10 (µg/m³)`	the two particle masses
`NOx index`	Sensirion's index, 1 = clean air
`Temperature (°C)`, `Humidity (%)`	from the BME280

A CSV saved with the app in English uses a comma between columns and a decimal point; in Romanian, French or German, a semicolon and a decimal comma, so Excel opens it correctly on a computer set to that language.
The particle sensor blocked. If something stays less than about 35 cm above the lens (a hand, a lid, a branch, a wall), the sensor gives no values and a yellow banner says **Particle sensor: no value. Nothing may stand within 35 cm above its lens (a lid, a hand, a wall). The PM columns stay empty while it is blocked.** A hand passing over it does no harm.
Cold. The particle sensor is meant for air above about +15 °C. The box warms it a little, but on a cold day its values may stop or drift: plan the walk for mild weather, and say so if you measure in the cold.

3. **At each place (about 7 min).** The holder keeps the box at about a metre above the ground, lens upwards, in the shade of their own body and with nothing above it. Press the place button and wait a minute or two for the temperature to settle. Then **Start recording**, five minutes on the stopwatch, **Stop recording**. Meanwhile one pupil notes what happens and when: a bus, a car waiting with its engine running, a mower, wind.
4. **On to the next place.** Choose the new place before **Start recording**; the rows of the walk all go into the file with their place code.
5. **Back in class (10 min).** Each group reads out what it noted at each place. Press **Export data** and save the workbook for the second lesson; for a first answer, the cards' typical values written on the pupil sheet are enough.

The investigation, in one table

The question. Where around our school does the air carry the most particles and nitrogen oxides, and where is it coolest?
The hypothesis. Written before the walk: the place we expect to be worst for each quantity, and why.
What I change. The place: P1 to P4.

What I measure. PM2.5, PM10, the NOx index, temperature and humidity, every 10 seconds for five minutes at each place, and what happens around us.
What I keep the same. The height of the box, the shade, the five minutes, the interval, nothing above the particle sensor.
How many repeats. The same route on another day or at another time of day. A difference that returns on the second walk is worth more than a large one seen once.
What makes a reading valid — and what makes it an artefact. Breath on the box raises the humidity and the temperature; direct sun on the box warms it; a hand or a branch over the lens stops the particle readings; someone smoking or vaping nearby gives a real but passing particle peak that belongs to that person, not to the place.

6. Analysis after the activity

6.1 What the data should look like

Quantity	What you usually see	Why
PM2.5	similar at the four places, with short peaks	Most of it is the town's background; local sources add peaks as they pass.
PM10	higher by the road or where dust is raised	Coarse dust settles close to its source.
NOx index	1 in quiet places; short rises by the road	The source is the exhaust of passing vehicles.
Temperature, humidity	warmer and drier over asphalt in sun, cooler and more humid by the trees	Absorbed sunlight against shade and evaporation.

In Excel. Sort the `Data` sheet by the `Place` column, select the PM2.5 cells of one place and read the **Average** in the status bar at the bottom of the window; write it down, then do the same for the next place and the next quantity. Older pupils also note the largest and smallest value at each place. **A difference between two places means something only if it is larger than the ups and downs within one place.** Expect PM2.5 from a few to a few tens of $\mu g/m^3$, and temperature differences of tenths of a degree to a few degrees.

6.2 Questions to ask afterwards

1. Which place had the highest mean PM10? Was it the one you predicted?
2. Is the difference in PM2.5 between the road and the green area larger than the rise and fall at either place?
3. The NOx index rose for a few seconds and then went back to 1. What happened at that moment?
4. The mean PM2.5 by the road was 20 $\mu g/m^3$. Can we say that the school exceeds the WHO guideline? Why not?
5. Where would you put a bench for pupils waiting to be collected, and why?

6.3 Assessment ideas

- **Low stakes:** the pupil fills in the table of the four places and names the warmest place and the dustiest.
- **Standard:** the pupil computes the means and explains where the particles and the nitrogen oxides at each place come from.
- **Extension:** a recommendation for the school (engines off at the entrance, a hedge along the fence) supported by the means, with a stated limitation of the measurement.

Common pitfalls and misconceptions

"The road is always the worst." For PM10 and the NOx index it often is; for PM2.5 the four places may differ by less than the readings jump at one place. That result is valid, and it teaches more than a confirmed guess.
Comparing five minutes with a 24-hour threshold. The WHO values are daily means. On a walk, compare the places with each other.
The NOx index read as a concentration. An index of 50 is not "50 micrograms of NO_2" but "more nitrogen oxides than this sensor has seen lately".
One reading instead of the mean. The cards change every second. Compare means of the rows, not the number on the screen at the moment a pupil looked.
Sun on the box. A box in direct sun reads too warm; keep it in the holder's shade at every place, including the green one.
An object above the particle sensor. A hand holding the box from above, an umbrella or a low branch stops the measurement. Hold the box from below and the sides.

Safety and classroom management

The road. P2 is chosen by the teacher, behind the fence or on the pavement several metres from the kerb. Nobody steps into the road or crosses it during the activity.
Exhausts and smoke. Never hold the box to an exhaust pipe or into smoke to "get a big number"; nobody stands behind a running engine.
Weather. The box is not waterproof: no measuring in rain. In cold weather keep the walk short.
The particle sensor's laser is enclosed in the sensor's housing and cannot be seen; even so, do not look closely into the lens and do not take the sensor apart.

Adaptations for pupils with special educational needs

One quantity per pupil. Pupils who tire quickly of tables follow a single quantity — for example only the temperature — and say at each place whether it is higher or lower than at the last. The class's conclusion is built from all of them.
Roles. One pupil holds the box, one times the five minutes, one notes events, one reads the cards aloud.
Noise and crowds. The entrance at the start or end of the day is loud and busy; choose a quieter time, or let a sensitive pupil wait with an adult at P3.
Mobility. Choose places reachable on paved paths; a pupil who cannot walk the route can keep the time and the notes at one place.

If things do not go to plan

The box has only one sensor. The lesson works with it. With temperature and humidity alone, the contrast between asphalt and grass is the whole lesson, and *Urban Heat Islands* goes further.

It rains. Measure at the open main door and at a window over the road, or move the walk to another day.
The laptop cannot come outside. Over WiFi the box works only within reach of the network; check the range from the yard beforehand, or carry the laptop on the USB cable.
The blocked-sensor warning does not go away. Look above the box up to about 35 cm: a hand, the lid of a bag, a branch, a wall. Move the box into the open.
The places ended up in separate files (the window was closed between two places; Stop and Start alone keep one file). Copy the rows of each file one under the other in a single sheet; the `Place` column keeps them apart.
The box will not connect. Use the "What the data should look like" table as a printed dataset and discuss it with the class.

7. Further resources

- **World Health Organization** — `https://www.who.int` — search for *WHO global air quality guidelines* (2021), with the values for PM2.5, PM10 and NO_2.
- **European Environment Agency** — `https://www.eea.europa.eu` — the European Air Quality Index map, hour by hour, including the stations in Romania.
- **Romania's national air quality monitoring network** — `https://www.calitateaer.ro` — the official station nearest the school, for the same hour as your walk.
- **Sensirion** — `https://sensirion.com` — search for the SGP41 and the note on its NOx index, which explains how the index follows the recent air.

8. Assessment

The exit ticket, identical to the one on the pupil sheet: *where around our school would you wait for someone to collect you? Give one of our measurements as the reason.*

For a more formal judgement, use the **common three-axis rubric**:

- **Measurement practice** — the pupil chooses the place before recording, holds the box the same way at every place, keeps the particle sensor clear and notes the events.
- **Data interpretation** — the pupil compares the means of the places, sets the differences against the spread within a place, and does not confuse five minutes with the WHO daily mean.
- **Scientific reasoning** — the pupil links each difference to a source (traffic, dust, shade) and states a limitation of the measurement: wind, time of day, a short sample.

The rubric itself, with descriptors for all levels, is defined in the *curriculum plan*.

9. Pupil sheet

The sheet is a separate document, in the folder `docs/en/student_sheets`: the file `breathing_around_the_school.pdf`. It fits on one sheet, printed on both sides, built on the three layers: what I see on the screen, what I do on my own, the conclusion I write. The table has a row for each of the four places and a spare row for a place of the class's own choosing; pupils leave empty the columns for sensors their box lacks.